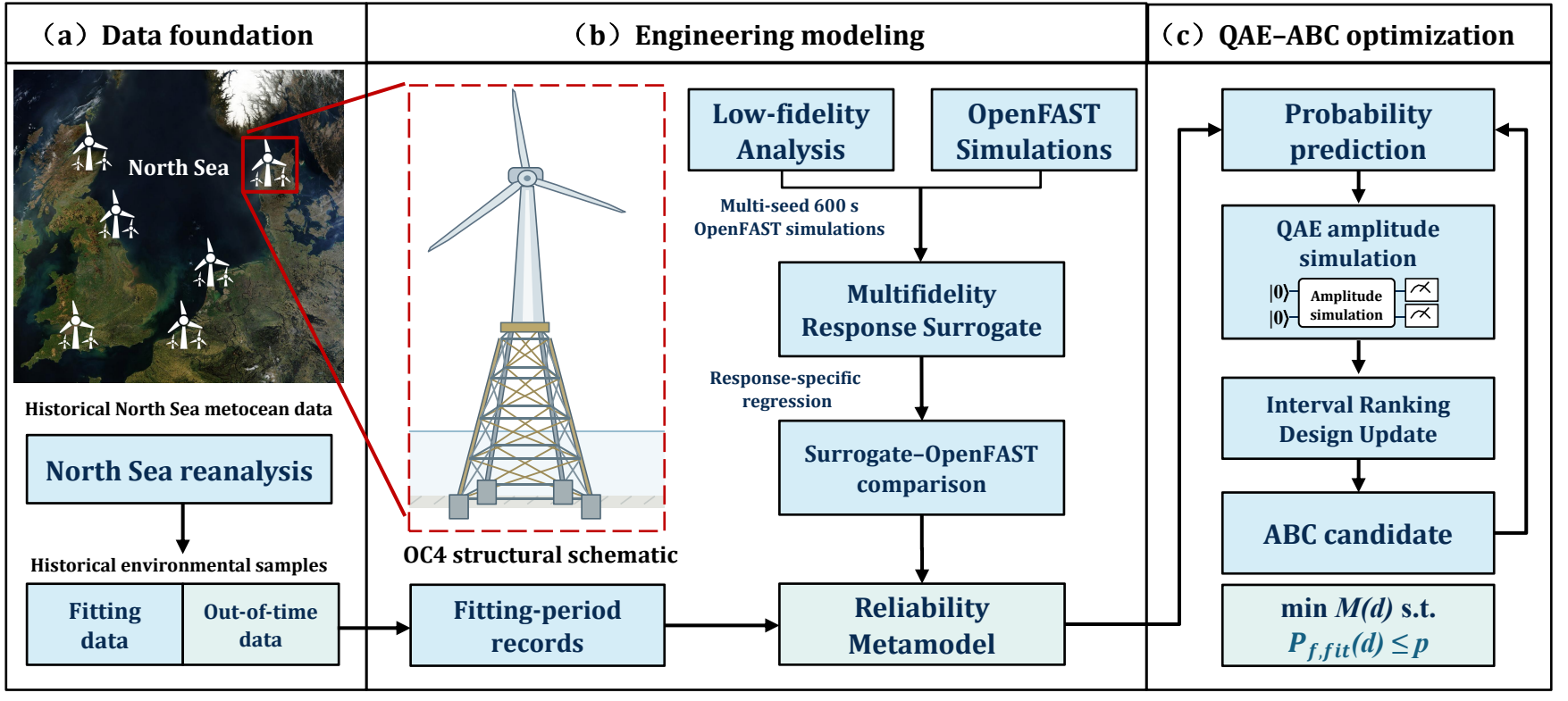

(a) Data foundation
North Sea
Historical North Sea metocean data
North Sea reanalysis
Historical environmental samples
Fitting data
Out-of-time data
OC4 structural schematic
(b) Engineering modeling
Low-fidelity Analysis
OpenFAST Simulations
Multi-seed 600 s OpenFAST simulations
Multifidelity Response Surrogate
Response-specific regression
Surrogate–OpenFAST comparison
Fitting-period records
Reliability Metamodel
(c) QAE–ABC optimization
Probability prediction
QAE amplitude simulation
|0⟩
|0⟩
Amplitude simulation
Interval Ranking Design Update
ABC candidate
min M(d) s.t.
P_f,fit(d) ≤ p